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LEARNER GUIDE

For

Data Visualisation with Tableau

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Conducted by

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17.0	15 August 2026	Rebuilt from legacy v16 coverage with concept-led slides, updated Tableau practice, ten self-contained labs and realistic synthetic Excel/Tableau assets.	Dr Alfred Ang

TABLE OF CONTENTS

How to Use This Guide	4
Learning Outcomes and Skills Framework	4
Skills Framework Criteria	5
Tableau Setup and File Discipline	5
Topic 01: Basic Tableau Features	6
Lab 1: Connect and Validate Retail Operations Data	6
Lab 2: Build Hierarchies and Drill Paths	8
Topic 02: Data Sources and Visualisation	10
Lab 3: Relate Orders, Products and Returns	10
Lab 4: Choose and Build Decision Charts	12
Topic 03: Data Transformation	14
Lab 5: Clean, Split and Pivot Campaign Data	14
Lab 6: Filter, Group and Set Customer Segments	16
Topic 04: Dashboards and Stories	18
Lab 7: Build an Executive Performance Dashboard	18
Lab 8: Add Actions and Tell a Performance Story	20
Topic 05: Calculations and Parameters	22
Lab 9: Create Dynamic Profitability Analysis	22
Topic 06: Analytics and Communication	24
Lab 10: Analyse Trends, Forecasts and Clusters	24
Assessment Flow	26
Support	26

How to Use This Guide

The slide deck teaches concepts, design choices and evidence interpretation. This Learner Guide holds the detailed click-by-click lab procedures. The ten labs form one connected Meridian Retail SG scenario and progress from source connection to dashboards, parameters and analytical interpretation.

Course and registration page: <https://www.tertiarycourses.com.sg/wsqa-data-visualisation-with-tableau.html>

Courseware and assessment portal: <https://lms-tms.tertiaryinfotech.com/>

Hands-on repository: <https://github.com/tertiarycourses/TGS-2020503177-Data-Visualisation-with-Tableau>

Learning Outcomes and Skills Framework

- LO1: Apply basic Tableau features for displaying information.
- LO2: Synthesise various plots to present data visually.
- LO3: Transform data to create informative and dynamic data displays.
- LO4: Create dashboards and scorecards to display internal and external benchmark data.
- LO5: Apply calculations and parameters to create interactive graphics, visuals and technical features.
- LO6: Perform analytics and communicate data limitations and interpretations of findings.

Skills Framework Criteria

Code	Type	Criterion
K1	Knowledge	Interpretation of data analysis and findings
K2	Knowledge	Types of information displays
K3	Knowledge	Suitability of data representations and visual displays for different contexts
K4	Knowledge	Data visualisation tools and techniques
K5	Knowledge	Elements of data dashboards
A1	Ability	Select appropriate visualisation techniques and information displays to convey data and findings
A2	Ability	Organise the presentation of data to reflect trends and correlations in a visually compelling way
A3	Ability	Incorporate appropriate elements to create informative and dynamic data displays
A4	Ability	Develop basic dashboards and scorecards to display internal and external benchmark data
A5	Ability	Incorporate interactive graphics, visuals and technical features into the data presentation
A6	Ability	Communicate limitations of data and interpretations of findings

Tableau Setup and File Discipline

- Use Tableau Desktop or Tableau Public. Tableau Public content may be public; never upload sensitive data.
- Open each lab from its own folder so the workbook can resolve the local Excel file.
- Save learner work as a packaged Tableau workbook (`.twbx`) so local data travels with the submission.
- Keep the original Excel source unchanged. Create transformations and calculations in Tableau, and reconcile to the Control sheet.
- Use accessible colour, visible filter state, truthful axes, readable labels and a named data period.
- Do not infer causality from visual association. State the population, filters, assumptions and material limitations.

Topic 01: Basic Tableau Features

Tableau ecosystem · interface · data roles · shelves · hierarchy

Key concepts

- **Visual analysis cycle:** Connect, explore, explain and refine around a decision question.
- **Dimensions and measures:** Dimensions slice the view; measures are aggregated to quantify it.
- **Discrete and continuous:** Blue headers create categories; green axes create a continuous range.
- **Workbook structure:** Worksheets build views; dashboards combine them; stories sequence insights.
- **Fields and metadata:** Names, types, roles, aliases and default properties shape every view.
- **Hierarchy:** Ordered levels let users drill from overview to operational detail.

Lab 1: Connect and Profile Service Operations Data

Alignment	LO1 · K1 · A1
Starter workbook	labs/lab-01-connect-and-profile-service-data/ lab-01-service-operations.xlsx
Tableau samples	sample/lab-01-service-profile.twb and sample/lab-01-service-profile.twbx
Expected output	A connected Tableau workbook with validated field roles and a service-volume view by region and channel.
Tools	Tableau Desktop/Public · Data Source page · Dimensions/Measures · Show Me

Business scenario. A customer-experience manager needs a first reliable view of service tickets across channels and regions.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Inspect the Scenario and Data Dictionary sheets Confirm the analytical grain: one row per service ticket.
2	Connect Tableau to the Tickets sheet Select the supplied Excel workbook and preview field types.
3	Correct data roles Set Opened Date to Date, Resolution Hours and Satisfaction Score to measures, and Ticket ID to a dimension.
4	Create folders Organise Customer, Service, Location and Outcome fields.
5	Build Service Volume by Region

Place Region on Rows, COUNTD(Ticket ID) on Columns and Channel on Colour.

6 Add useful context

Show the record count, date range and a concise tooltip.

7 Reconcile the view

Compare Tableau's total to the Excel Control sheet.

8 Save evidence

Save a packaged workbook and capture the final view.

Acceptance test

The workbook connects without broken fields; dates and numbers have correct roles; the view reconciles to the Excel record count.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Lab 2: Build Hierarchies and Drill Paths

Alignment	LO1 · K1 · A1
Starter workbook	labs/lab-02-build-hierarchies-and-drill-paths/ lab-02-retail-hierarchy.xlsx
Tableau samples	sample/lab-02-hierarchy.twb and sample/lab-02-hierarchy.twbx
Expected output	A geographic and product hierarchy with a drillable sales-and-margin view.
Tools	Folders · Hierarchies · Drill up/down · aliases · default properties

Business scenario. Retail leaders need to move from national performance to region, outlet and department detail without losing context.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Connect to the Retail_Data table Review geography and product levels before building the view.
2	Create Geography hierarchy Region > District > Outlet Name.
3	Create Product hierarchy Category > Subcategory > Product Name.
4	Create a folder structure Group identifiers, geography, product, commercial measures and controls.
5	Build Sales and Profit by Geography Use the Geography hierarchy and drill one level at a time.
6	Add Profit Ratio Use the supplied Profit and Sales measures without changing the source grain.
7	Test drill behaviour Verify totals remain stable while lower levels reveal detail.
8	Document the hierarchy purpose State who uses each drill path and what decision it supports.

Acceptance test

Both hierarchies drill in the correct order and totals remain stable at every displayed level.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Topic 02: Data Sources and Visualisation

Connections · extracts · relationships · joins · blending · chart choice

Key concepts

- **Connection strategy:** Choose live or extract by freshness, performance and governance needs.
- **Relationships:** Preserve each table's grain and combine data at analysis time.
- **Joins:** Physically combine rows; cardinality and unmatched keys determine the result.
- **Visual encoding:** Position and length usually support more accurate comparisons than area or colour.
- **Chart purpose:** Use bars for comparison, lines for time, scatterplots for relationship and treemaps for hierarchy.
- **Aggregation:** SUM, AVG, COUNT and distinct count answer different business questions.

Lab 3: Relate Orders, Products and Returns

Alignment	LO2 · K3 · A2
Starter workbook	labs/lab-03-relate-orders-products-and-returns/ lab-03-related-retail-data.xlsx
Tableau samples	sample/lab-03-relationships.twb and sample/lab-03-relationships.twbx
Expected output	A governed multi-table data model plus an extract and reconciliation sheet.
Tools	Relationships · joins · cardinality · referential integrity · extract

Business scenario. Meridian Retail SG stores orders, product master data, outlets and returns at different grains.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Review the four source tables Orders, Products, Outlets and Returns have different grains.
2	Relate Orders to Products Use Product ID with many Orders to one Product.
3	Relate Orders to Outlets Use Outlet ID with many Orders to one Outlet.
4	Relate Orders to Returns Use Order ID and preserve the order grain.

5	Check unmatched keys Use the Control sheet and a Tableau null check.
6	Compare with an inner join Observe how unmatched rows and duplicated rows would affect totals.
7	Create an extract Save a local extract and run a refresh.
8	Reconcile Confirm Sales and Order Count match the Excel controls.

Acceptance test

Order totals are not duplicated, unmatched keys are visible, and the extract refreshes from the supplied workbook.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Lab 4: Choose and Build Decision Charts

Alignment	LO2 · K3 · A2
Starter workbook	labs/lab-04-choose-and-build-decision-charts/ lab-04-chart-selection.xlsx
Tableau samples	sample/lab-04-decision-charts.twb and sample/lab-04-decision-charts.twbx
Expected output	Four fit-for-purpose worksheets: ranked bar, quarterly line, profit-sales scatterplot and category treemap.
Tools	Bar · line · scatterplot · treemap · labels · tooltips · sort

Business scenario. A commercial director needs comparison, trend, relationship and portfolio views for one monthly review.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Read the four management questions Identify comparison, trend, relationship and composition purposes.
2	Create a ranked bar chart Sales by Market, coloured by Category.
3	Create a quarterly line chart Sales by Order Date quarter, coloured by Category.
4	Create a scatterplot Sales versus Profit by Outlet with Size by Orders.
5	Create a treemap Sales by Category and Subcategory.
6	Remove chart clutter Use truthful axes, readable labels and focused tooltips.
7	Annotate one insight per view Separate what is observed from why it may have occurred.
8	Validate and save Reconcile displayed values to the Control sheet.

Acceptance test

Every chart answers its stated question, uses truthful axes, and includes enough context to interpret the marks.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Topic 03: Data Transformation

Data Interpreter · split · merge · pivot · filters · groups · sets

Key concepts

- **Analysis-ready structure:** One row is one observation; one column is one variable; one table is one grain.
- **Data Interpreter:** Removes presentation clutter from spreadsheets while preserving the underlying evidence.
- **Split and pivot:** Separate compound fields and reshape wide columns into analysis-ready rows.
- **Filter order:** Extract, source, context, dimension and measure filters do not behave identically.
- **Groups:** Combine members into a stable business category.
- **Sets:** Define a dynamic in/out population that can drive comparisons and actions.

Lab 5: Clean, Split and Pivot Campaign Data

Alignment	LO3 · K4 · A3
Starter workbook	labs/lab-05-clean-split-and-pivot-campaign-data/ lab-05-messy-campaign-data.xlsx
Tableau samples	sample/lab-05-transformation.twb and sample/lab-05-transformation.twbx
Expected output	An analysis-ready long table with clean location fields and documented transformation checks.
Tools	Data Interpreter · custom split · merge · pivot · aliases

Business scenario. A campaign export contains banner rows, compound location codes and monthly measures spread across columns.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Open the messy workbook Note banner rows, merged cells, compound fields and monthly columns.
2	Use Data Interpreter Select the detected campaign table and verify headers.
3	Split Location Code Separate Region and District with a custom split on .
4	Create a clean Campaign Name Trim spaces and standardise aliases.

5	Pivot monthly columns Pivot Jan 2025 through Dec 2025 into Month and Spend.
6	Set field types Month is Date; Spend and Conversions are measures.
7	Filter unusable records Exclude only documented blank campaign keys.
8	Reconcile transformation Match the long-table spend total to the Excel Control sheet.

Acceptance test

The transformed data has one row per campaign-month, clean location fields, no blank campaign key and reconciled spend totals.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Lab 6: Filter, Group and Set Customer Segments

Alignment	LO3 · K4 · A3
Starter workbook	labs/lab-06-filter-group-and-set-customer-segments/lab-06-customer-segments.xlsx
Tableau samples	sample/lab-06-groups-sets-filters.twb and sample/lab-06-groups-sets-filters.twbx
Expected output	A segment view using context filters, grouped categories and a top-customer set.
Tools	Dimension/measure/date filters · context filter · groups · sets

Business scenario. The loyalty team wants a stable business grouping plus a dynamic high-value customer cohort.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Connect to Customer_Orders Review customer, period, segment and profitability fields.
2	Create Business Segment group Combine Micro and Small into SME; retain Mid-Market and Enterprise.
3	Create a date filter Limit analysis to a selected rolling period.
4	Create a context filter Make the selected Region the context for Top N analysis.
5	Create High-Value Customer set Top 20 customers by Sales within the context.
6	Compare IN versus OUT Show Profit Ratio and Order Frequency by set membership.
7	Add a measure filter Exclude customers with non-material sales only when justified.
8	Explain the cohort Document how filter order affects membership.

Acceptance test

The high-value set responds to the selected period and the business grouping is explainable from the source fields.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Topic 04: Dashboards and Stories

KPI scorecards · layout · actions · device design · story sequence

Key concepts

- **Four Ws:** Who is the audience, what decision, when is it used and where is action taken?
- **Dashboard hierarchy:** Lead with decision KPIs, then trends, drivers and diagnostic detail.
- **Layout containers:** Tiled containers create predictable alignment; floating objects need deliberate control.
- **Actions:** Filter, highlight, URL, navigation, parameter and set actions create purposeful interaction.
- **Benchmark context:** Targets, prior periods and peer ranges turn values into performance signals.
- **Stories:** A sequence of story points communicates change, evidence and decision.

Lab 7: Build an Executive Performance Dashboard

Alignment	LO4 · K5 · A4
Starter workbook	labs/lab-07-build-executive-performance-dashboard/lab-07-dashboard-scorecard.xlsx
Tableau samples	sample/lab-07-executive-dashboard.twb and sample/lab-07-executive-dashboard.twbx
Expected output	A 16:9 scorecard dashboard with KPI tiles, trend, ranked outlets, benchmark context and visible filter state.
Tools	Dashboards · layout containers · KPI sheets · device preview · captions

Business scenario. Executives need one screen to monitor sales, margin, returns and target attainment across outlets.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Define the executive decisions Monitor performance, locate exceptions and assign follow-up.
2	Build KPI sheets Sales, Profit, Profit Ratio, Return Rate and Target Attainment.
3	Build the trend view Monthly Actual versus Target with clear reference context.
4	Build ranked outlets Sort outlets by target variance and highlight material exceptions.

5	Assemble a 16:9 dashboard Use tiled containers and a clear visual hierarchy.
6	Add filters Region, Channel and Month with visible filter state.
7	Test device layout Confirm the presentation view remains readable.
8	Run the acceptance checklist Reconcile totals and remove every object without a decision purpose.

Acceptance test

The dashboard is readable at presentation size, totals reconcile, active filters are visible and every element supports a named decision.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Lab 8: Add Actions and Tell a Performance Story

Alignment	LO4 · K5 · A4
Starter workbook	labs/lab-08-add-actions-and-tell-a-performance-story/lab-08-actions-story.xlsx
Tableau samples	sample/lab-08-actions-story.twb and sample/lab-08-actions-story.twbx
Expected output	A dashboard with filter/highlight/navigation actions and a five-point performance story.
Tools	Filter action · highlight action · navigation · URL action · story points

Business scenario. Regional managers need to explore a dashboard and then present a concise evidence sequence to leadership.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Open the base performance views Identify which sheets are sources and which are targets.
2	Add a filter action Select a Region to filter outlet and category detail.
3	Add a highlight action Hover or select a Category to highlight across related views.
4	Add a navigation action Move from executive summary to diagnostic detail.
5	Add a safe URL action Open the supplied internal-style reference URL field only.
6	Create five story points Situation, trend, exception, driver and recommended action.
7	Test reset behaviour Users must be able to return to the full population.
8	Record an action statement Name owner, next step, trigger and review date.

Acceptance test

Actions work from the intended source sheets, preserve context and the story ends with an evidence-backed action.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
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Topic 05: Calculations and Parameters

Calculated fields · table calculations · logic · parameters · dynamic views

Key concepts

- **Row-level calculation:** Evaluated for each record before aggregation.
- **Aggregate calculation:** Combines measures at the view's level of detail.
- **Table calculation:** Computed over the marks already returned to the visualisation.
- **Logical design:** IF, IIF and CASE translate business rules into transparent classifications.
- **Parameters:** Single values controlled by the user; pair them with calculations or actions to change behaviour.
- **Dynamic view:** A parameter can switch metrics, thresholds, reference values or dimensions.

Lab 9: Create Dynamic Profitability Analysis

Alignment	LO5 · K2 · A5
Starter workbook	labs/lab-09-create-dynamic-profitability-analysis/ lab-09-calculations-parameters.xlsx
Tableau samples	sample/lab-09-dynamic-profitability.twb and sample/lab-09-dynamic-profitability.twbx
Expected output	A dynamic metric view with calculated fields, parameter controls, table calculations and a threshold filter.
Tools	Calculated fields · IF/CASE · table calculations · parameters · dynamic titles

Business scenario. Finance wants to switch between sales, profit and margin, and test which products exceed a user-defined threshold.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Review supplied controls Excel includes source totals and threshold examples.
2	Create Profit Ratio $SUM([Profit]) / SUM([Sales])$ with a zero-safe condition.
3	Create Selected Metric parameter Sales, Profit, Profit Ratio and Orders.
4	Create Dynamic Metric calculation Use CASE to return the selected measure.

5	Create Sales Threshold parameter Integer range 0 to 85000 with step 100.
6	Create threshold filter Keep products where SUM([Sales]) is at least the parameter.
7	Add Percent of Total Use a table calculation with explicit addressing.
8	Create a dynamic title and test The title and marks must respond to both controls.

Acceptance test

Metric and threshold controls change the view correctly, formulas reconcile to Excel controls and invalid denominator cases are handled.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Topic 06: Analytics and Communication

Reference lines · trends · forecasts · clusters · limitations · data stories

Key concepts

- **Reference context:** Average, median, target and distribution bands help the audience judge performance.
- **Trend models:** A fitted line summarises association; it does not establish causality.
- **Forecast:** Tableau extends a time series under model assumptions and expresses uncertainty.
- **Clustering:** Groups similar marks from selected features; business labels require interpretation.
- **Limitations:** Data coverage, quality, granularity, bias and model assumptions bound the conclusion.
- **Responsible narrative:** Separate observation, interpretation, recommendation and uncertainty.

Lab 10: Analyse Trends, Forecasts and Clusters

Alignment	LO6 · K1 · A6
Starter workbook	labs/lab-10-analyse-trends-forecast-and-clusters/ lab-10-analytics-communication.xlsx
Tableau samples	sample/lab-10-analytics-story.twb and sample/lab-10-analytics-story.twbx
Expected output	An analytics workbook with reference context, trend, forecast, clustering and a limitations-led decision narrative.
Tools	Reference lines · trend lines · forecast · clustering · describe trend/model

Business scenario. The strategy team needs a cautious outlook for outlet demand and a segmentation of store operating patterns.

Before you begin

- Install Tableau Desktop or Tableau Public and confirm that local Excel connections work.
- Copy the lab folder to a writable location and keep the Excel file, `.twb` and `.twbx` together.
- Start with the root Excel workbook. Use the sample only after attempting the activity.
- Treat all records as synthetic training data; never replace them with personal or live company data.

Detailed procedure

1	Build the monthly outlet trend Use at least 24 complete months and show the observed period.
2	Add average and target lines Label the benchmark and avoid ambiguous colours.
3	Add a trend line Describe slope, fit and whether the relationship is only associative.
4	Create a forecast Show prediction intervals and document horizon and seasonality assumptions.

5	Create clusters Use Sales, Profit Ratio, Return Rate and Footfall; inspect cluster summaries.
6	Name clusters cautiously Use business labels only after reviewing the feature profiles.
7	Write limitations Cover missing drivers, aggregation, selection bias and changing conditions.
8	Publish the decision narrative State observation, interpretation, recommendation, uncertainty and monitoring trigger.

Acceptance test

The workbook states assumptions and limitations, distinguishes association from causation, and reports at least one action with a monitoring trigger.

Evidence to retain

- Packaged Tableau workbook (`.twbx`) with the required sheets, dashboard or story.
- Screenshot of the principal output at readable presentation size with the active filter state visible.
- Reconciliation against the Excel Control sheet: record count plus relevant sales, profit, spend or service metric.
- Interpretation note that separates observation, business meaning, limitation and recommended next action.

Common pitfalls and recovery

- Broken Excel path: Edit the Tableau data connection and point it to the workbook in the same lab folder.
- Unexpected totals: return to the Data Source page and check relationship/join keys, duplicate rows and filters.
- Wrong chart structure: inspect whether a field is a dimension or measure and whether it is discrete or continuous.
- Slow or confusing dashboard: reduce unnecessary marks, quick filters and objects; retain only decision-relevant interaction.

Assessment Flow

- Complete TRAQOM / SSG digital attendance.
- Complete assessment digital attendance.
- Sit the 60-minute Written Assessment followed by the 60-minute Practical Performance assessment.
- Upload the completed question papers and required Tableau evidence to the LMS.
- Sign the Assessment Summary Record.

LMS: <https://lms-tms.tertiaryinfotech.com/>

Support

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